

SOLOMON

Solomon Ai Solutions & Application for Semi-Conductor Industry

Breaking Limits: Powering the AI Era.



**SEMICON[®]
TAIWAN**

Partial Customer List - US & Europe






















Partial Customer List - Asia



















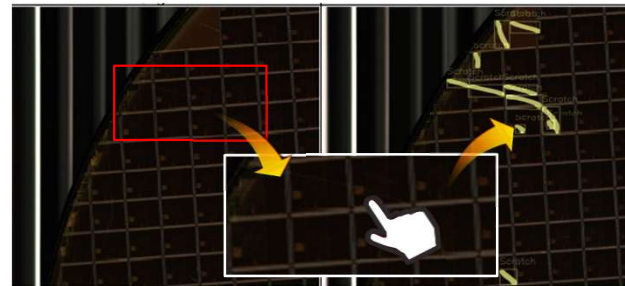
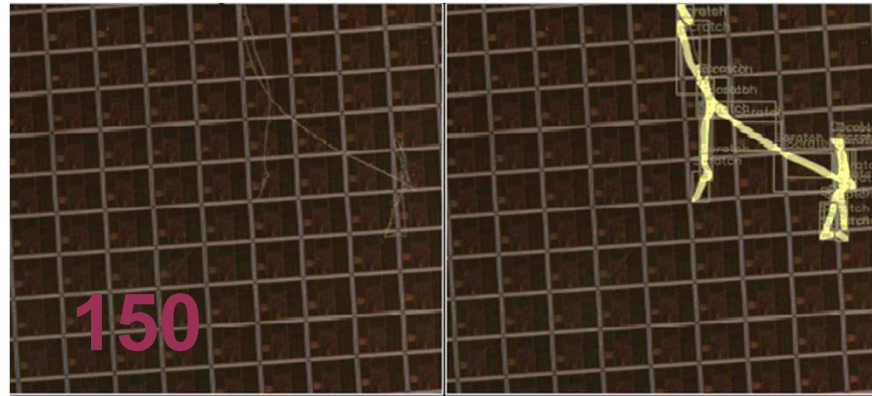
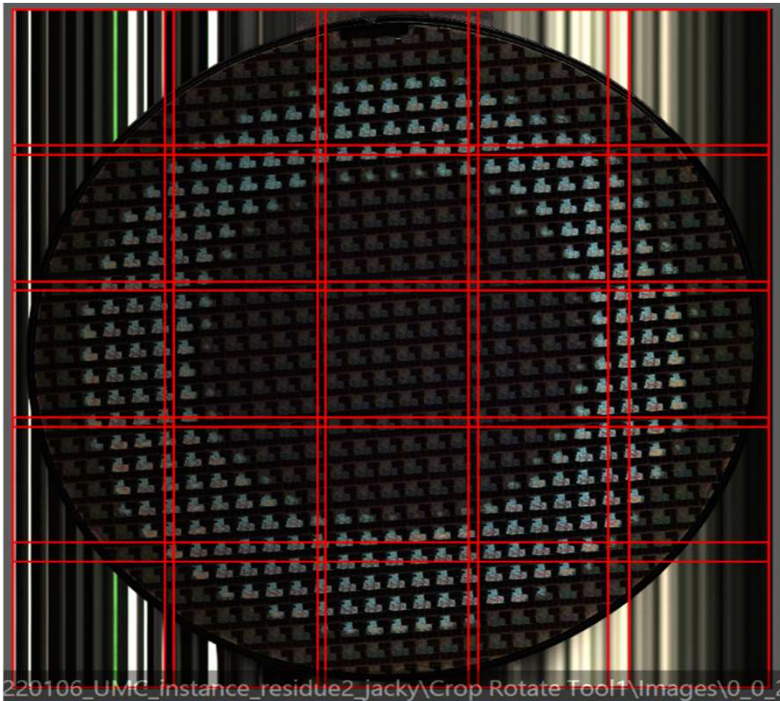




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Wafer Surface Inspection

- Using cutting techniques to segment the wafer, inspecting for micro defects.
- Easily overcomes slight scratches that are difficult to detect with the naked eye.

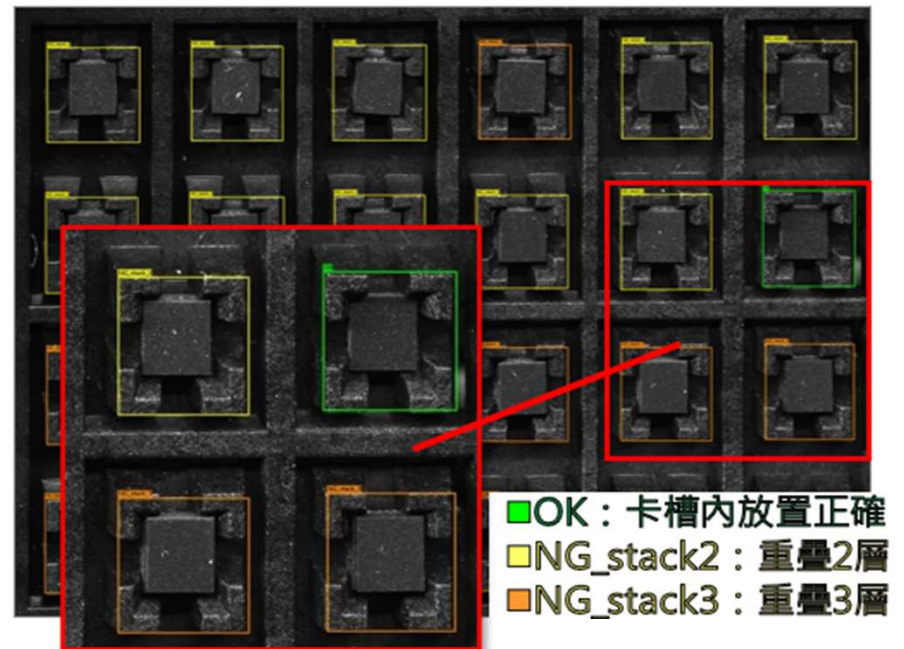
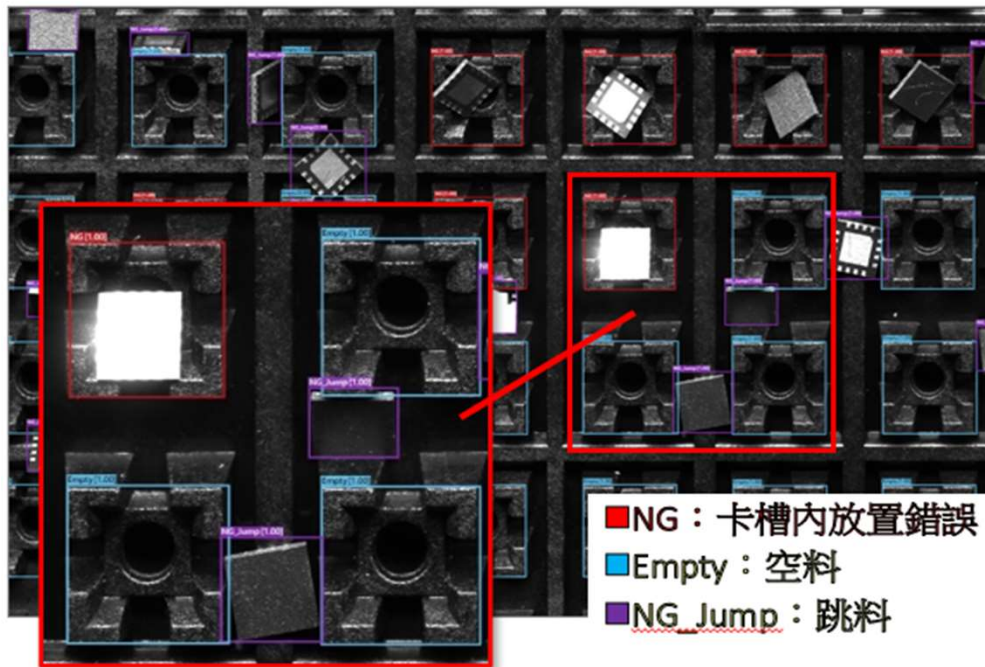


Scratches are not obvious and require magnification for inspection.

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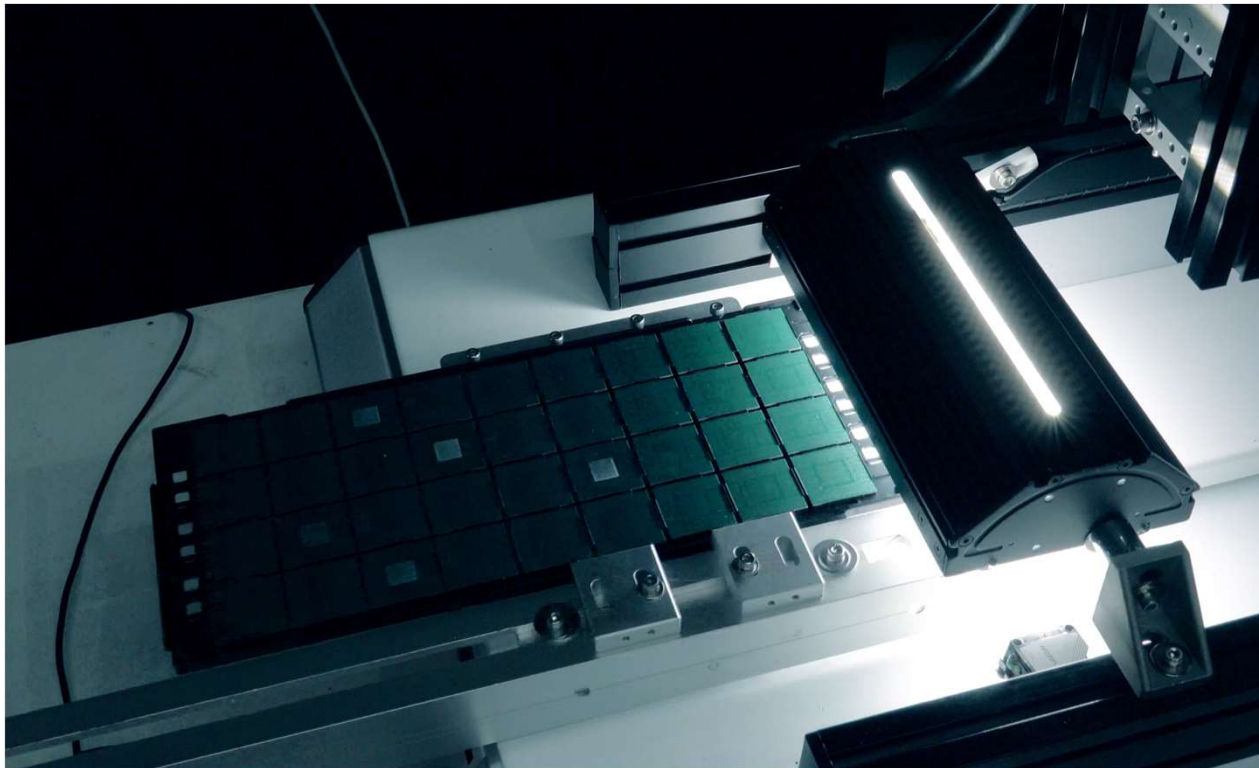
IC Components on the Tray

- Inspecting the Tray for Abnormal IC Placement
- IC Overlap, Difficult to Detect with the Naked Eye



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Line scan camera inspection of IC package substrates



- Integrating the advantages of AOI and AI inspection, this solution addresses the issue of overkills (false positives) in traditional AOI equipment caused by stringent conditions.
- The system can define various types of defects and accurately mark their locations on the tray, significantly reducing the cost of damage handling.

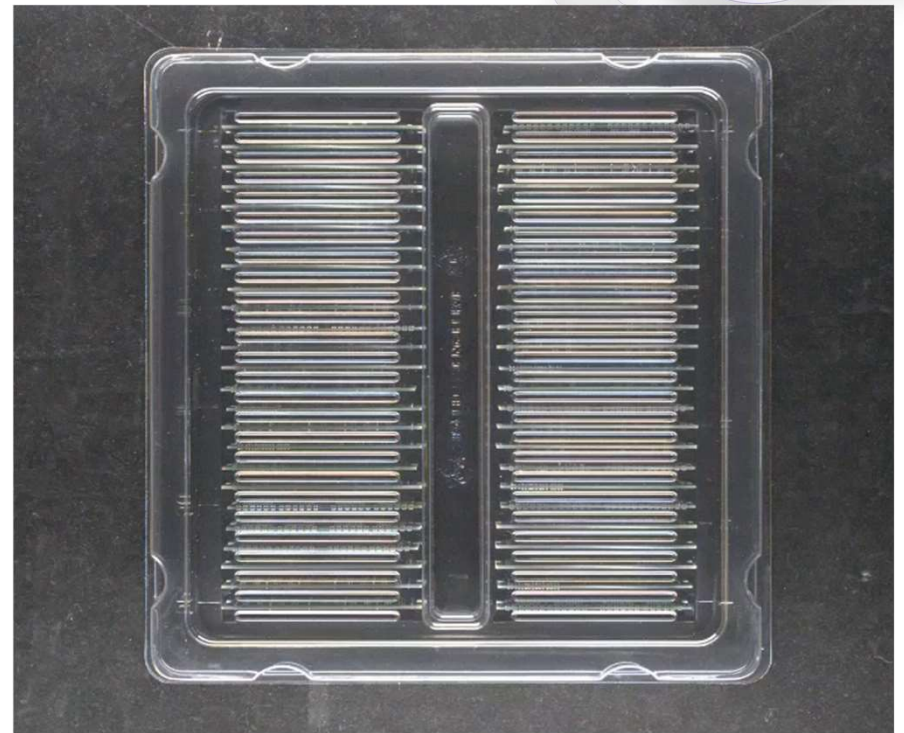
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Memory Counting Under Blister Covers

- Reflective or transparent materials typically lead to detection errors or make detection difficult.
- By detecting subtle changes in color gradients, memory units can be accurately counted.



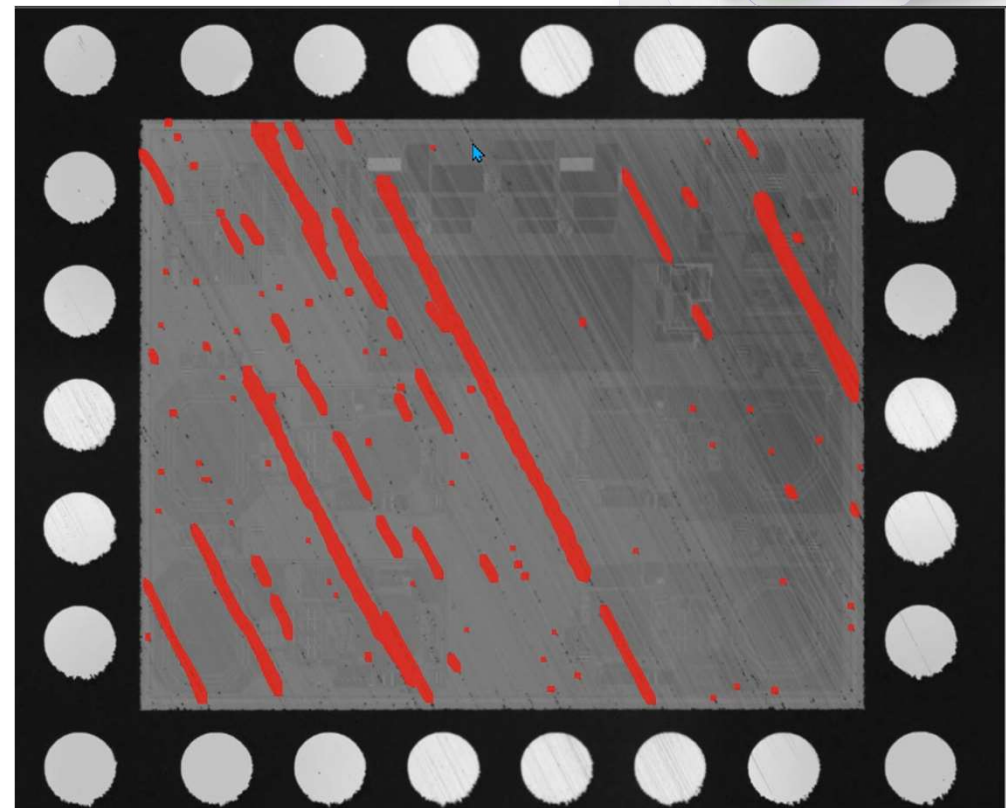
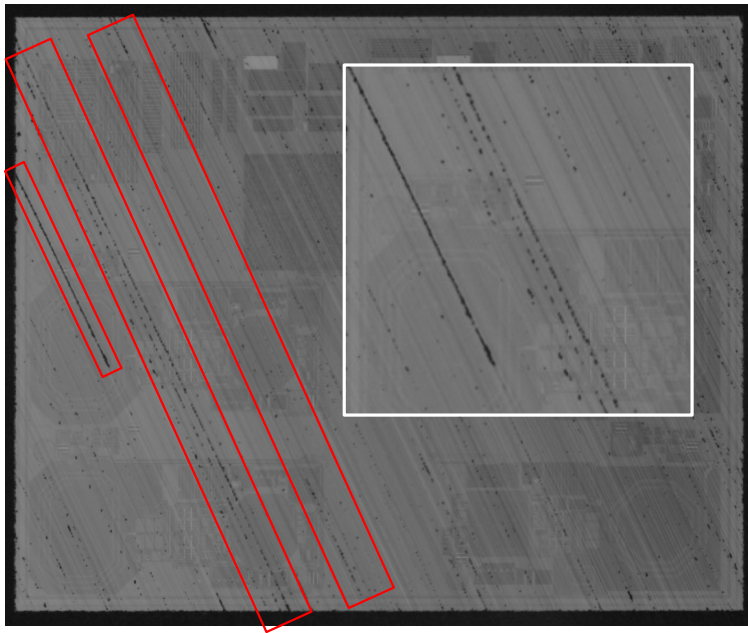
Applicable to reflective or translucent materials.



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IC Inspection for Grinding & Contamination

- To find all the grinding defect, scratch, and contamination, including all size of defects.
- The background also have some line mark similar to the defect feature.

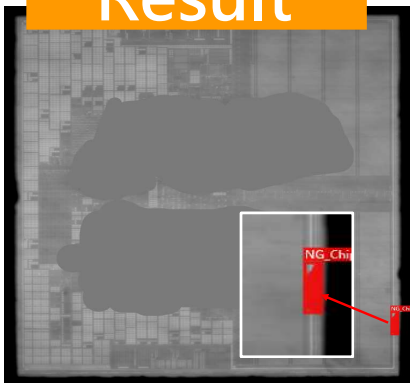


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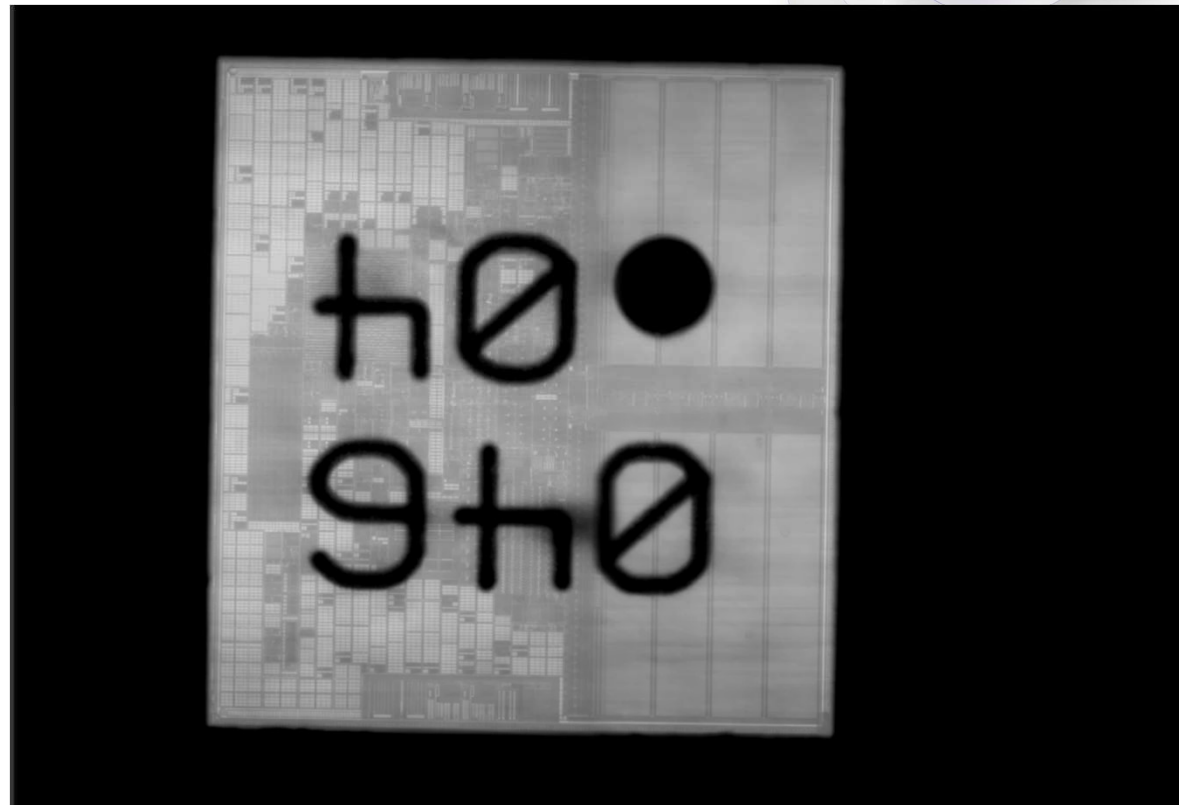
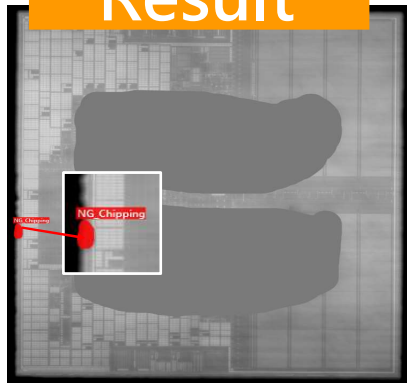
Micro-cracks Inspection after Die-saw

- Inspect the micro-crack through AI learning, to solve the low contrast problem.
- Complex background design with different SKU could also affect the inspection.

Result



Result

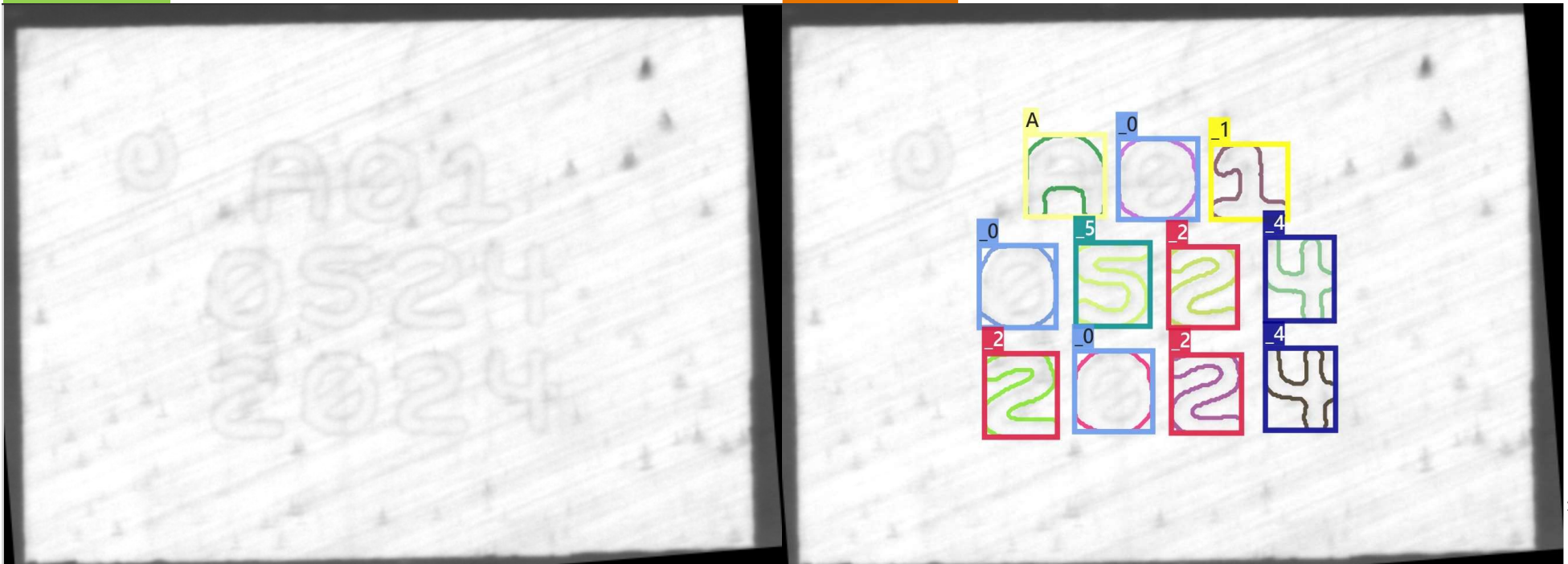


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Laser Marking OCR

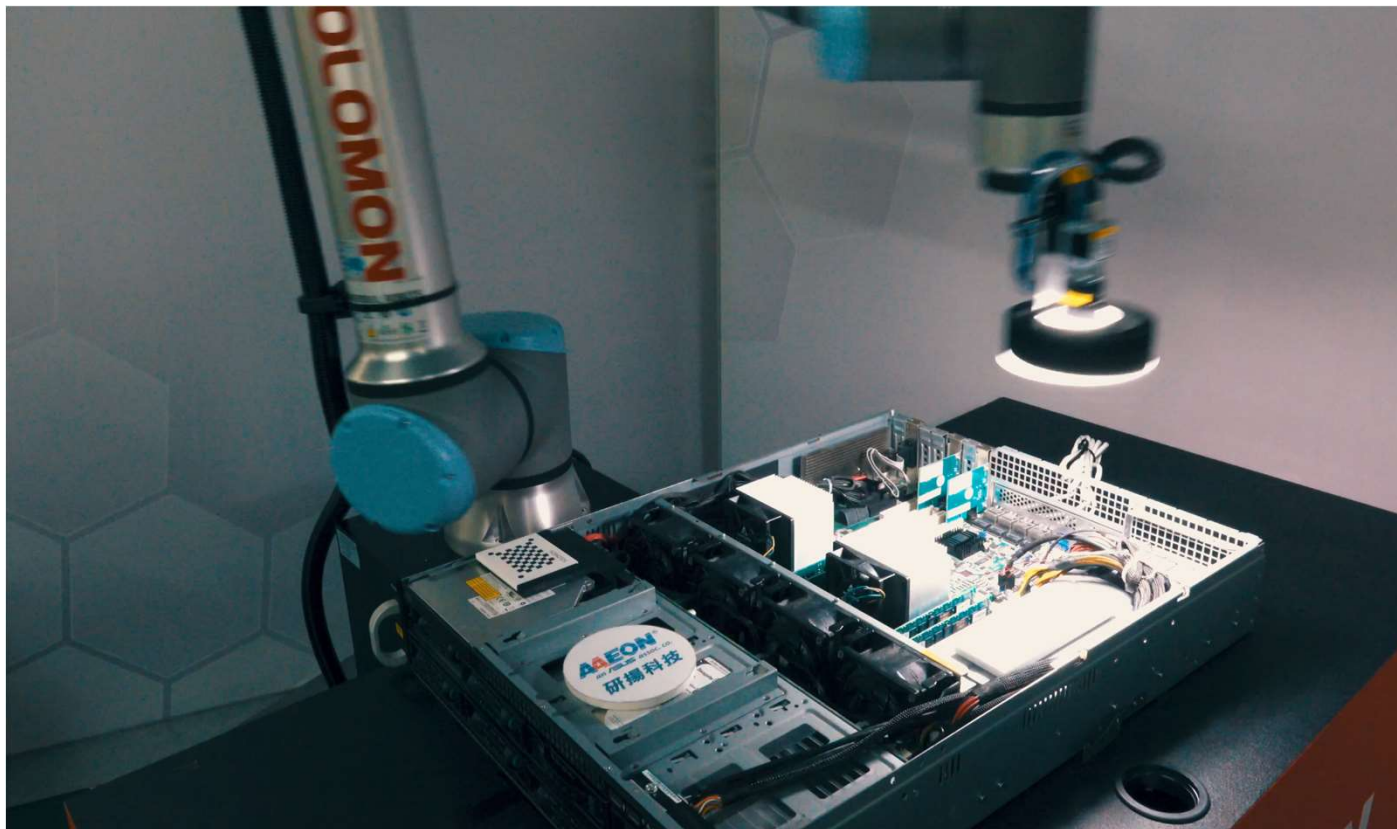
Original

OCR



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Robotic arm contoured object AI inspection



- A robotic arm equipped with Solomon's AI recognition technology and various 2D/3D cameras.
- Autonomously adjusts its path based on the object's shifted position, quickly moves to the capture point.
- It takes photos to instantly detect common production line quality issues such as defects and missing components.

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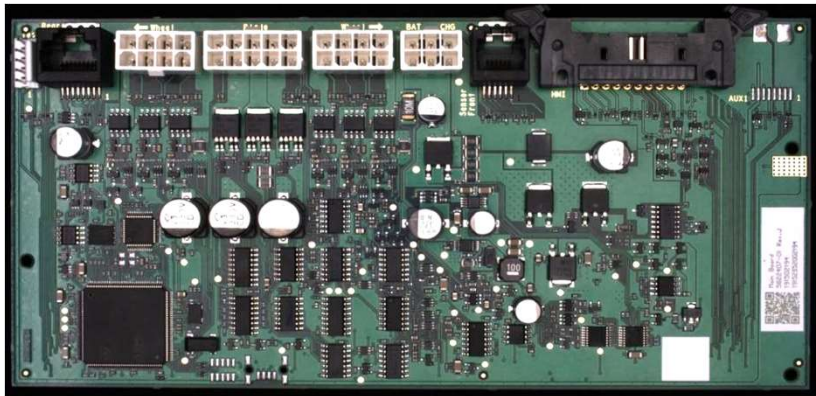
PCB Defect Detection (Lithuania)

■ Electronics Manufacturing

- Detection of defects such as open circuits, short circuits, plating defects, residual copper, and missing components.

■ Solutions and Benefits

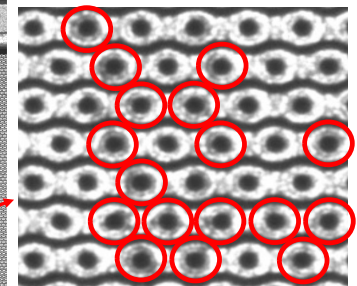
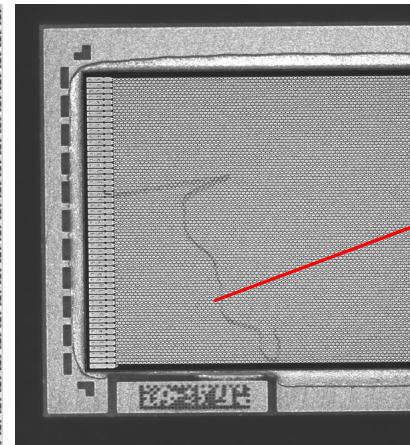
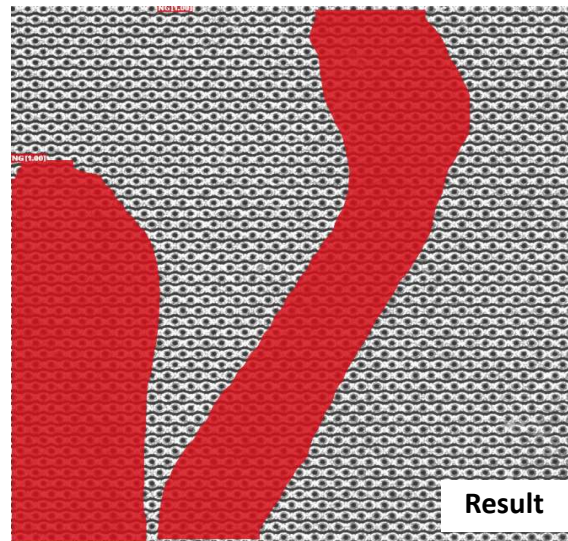
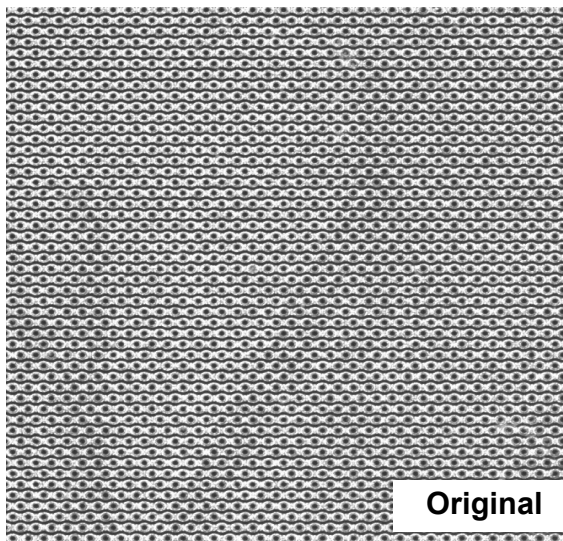
- Overcoming the impact of product variations to detect PCB defects.
- Reducing the cost and errors associated with manual inspection.
- Providing UI integration with cameras and PLCs.



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Automotive Circuit Board Defect Detection

- **Client Facility Location:** Singapore
- **Client Industry:** Automotive Sensor Manufacturing
- **Client Requirements:**
 - Detection of light scratches that are difficult to identify.
 - Detection of defects with unclear boundaries.
- **Benefit:** Significantly reduces the misjudgment rate of automotive circuit boards.



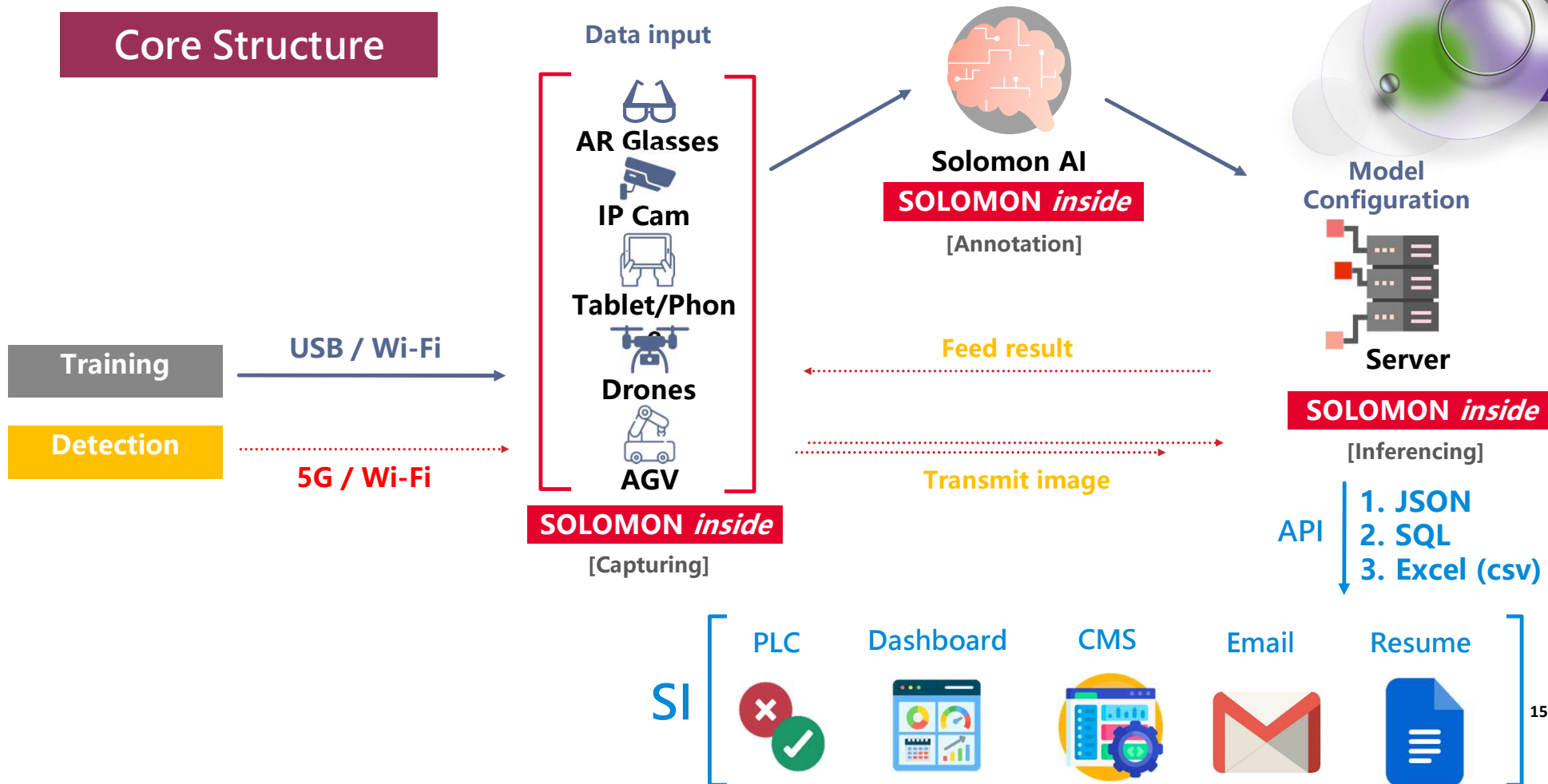
META-aivi

META-aivi Smart Device Compatibility

Solomon's AI-p 所羅門的AI+AR平台可以與多種終端設備無縫整合 smart devices

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Core Structure



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Automation vs Human



Speed up operation time

Optimizing complex steps by digitalizing operations.



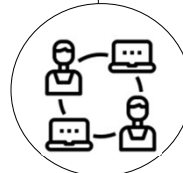
Shorten new staff training

AR guided step-by-step instructions can reduce employee learning costs.



Increase workplace safety

Using AI to alert workers in order to avoid incorrect operation and human error.



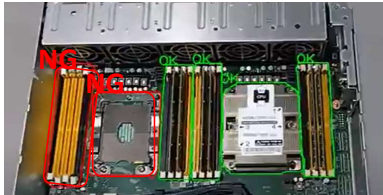
Instant remote support

Instant operators support through live communication, helping to overcome geographical limitations.

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Server Assembly

- Solomon AI can quickly identify **missing components**, displaying results on AR glasses in real-time. With AI guidance, operators can **align** components correctly, conduct inspection and obtain final product information.



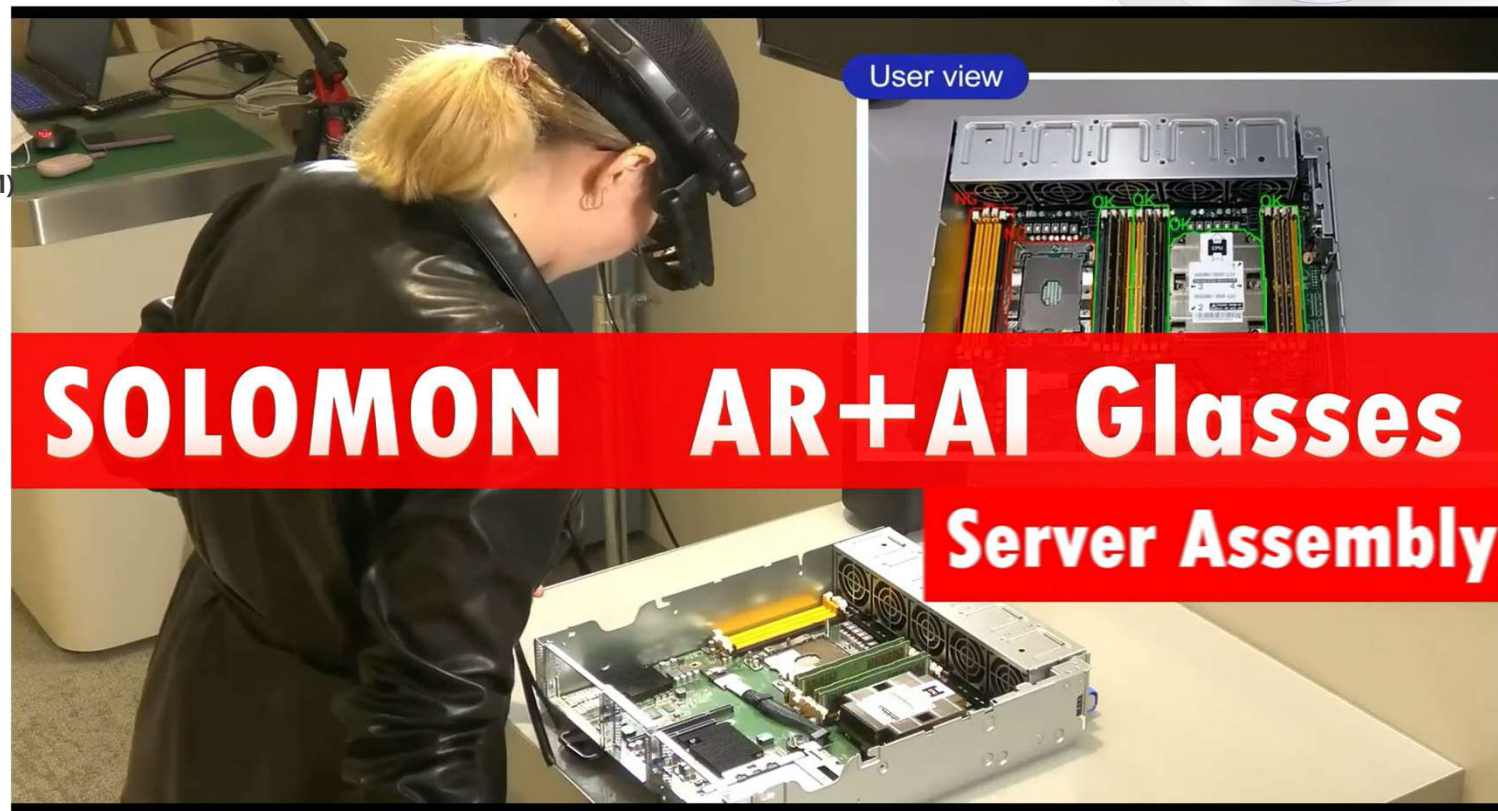
1. AI Inspection (Components: CPU, RAM)



2. **Instant** results display



3. AI verification **completed**



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PCB board connector assembly SOP



➔ Using Solomon AI to prompt assembly personnel to follow the **assembly SOP for PCB** board connectors on the assembly line. This ensures that employees¹⁸ can assemble according to instructions and verify the results afterwards to ensure production quality.

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Microscopic AI Intelligent Inspection and Guidance System

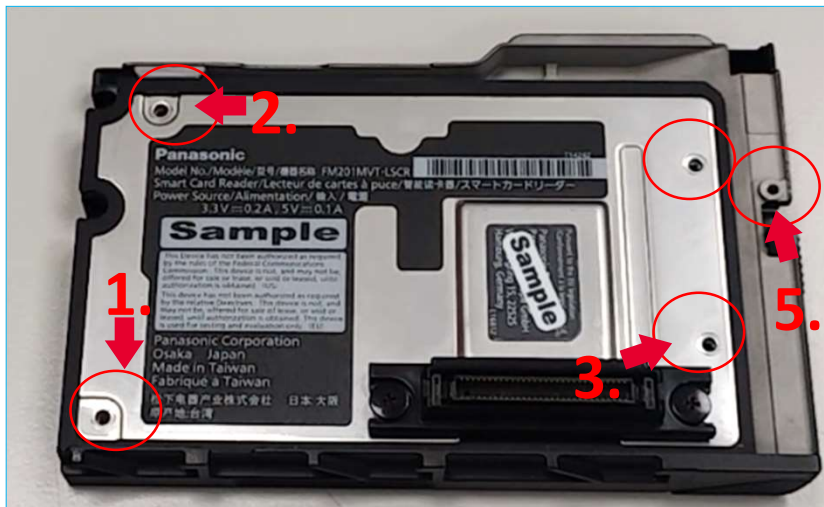


- Using various microscopic cameras to detect tiny defects or differences, this system is compatible with different devices.
- It leverages AI to assist personnel in quality inspection or assembly guidance while also maintaining an electronic record.

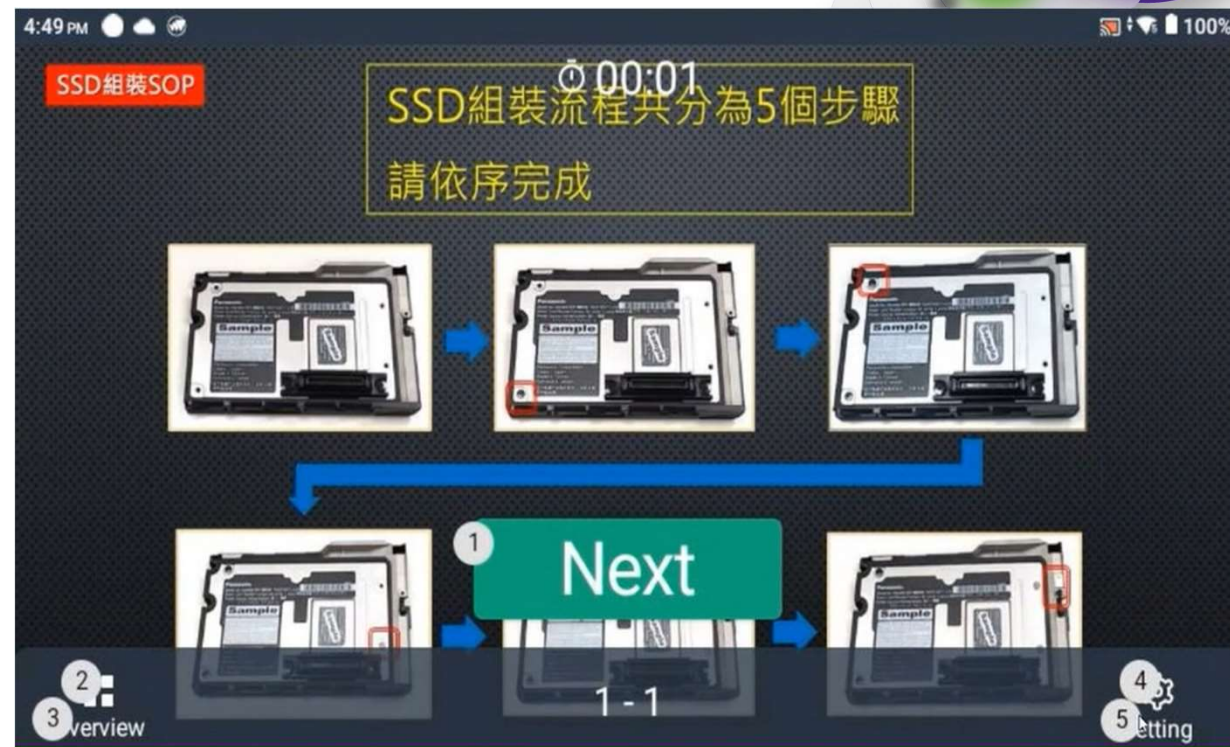


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SSD Assembly SOP Validation



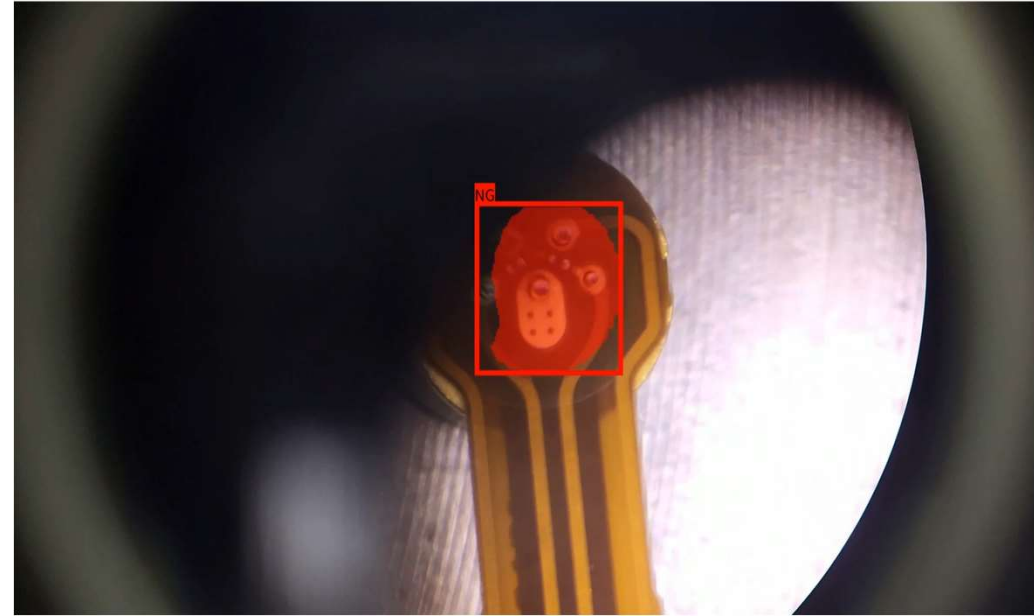
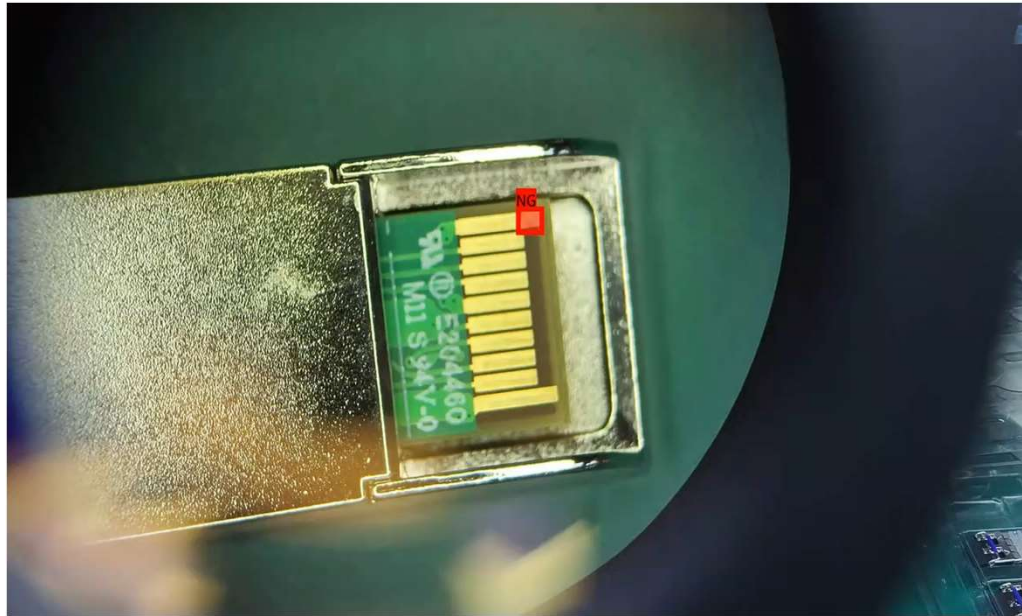
According to the **position of the screws** on the SSD, place and tighten the screws **in order from 1 to 5**.



➡ Solomon AI **guide the operator through the assembly process** and detects whether the screws are placed and tightened correctly.

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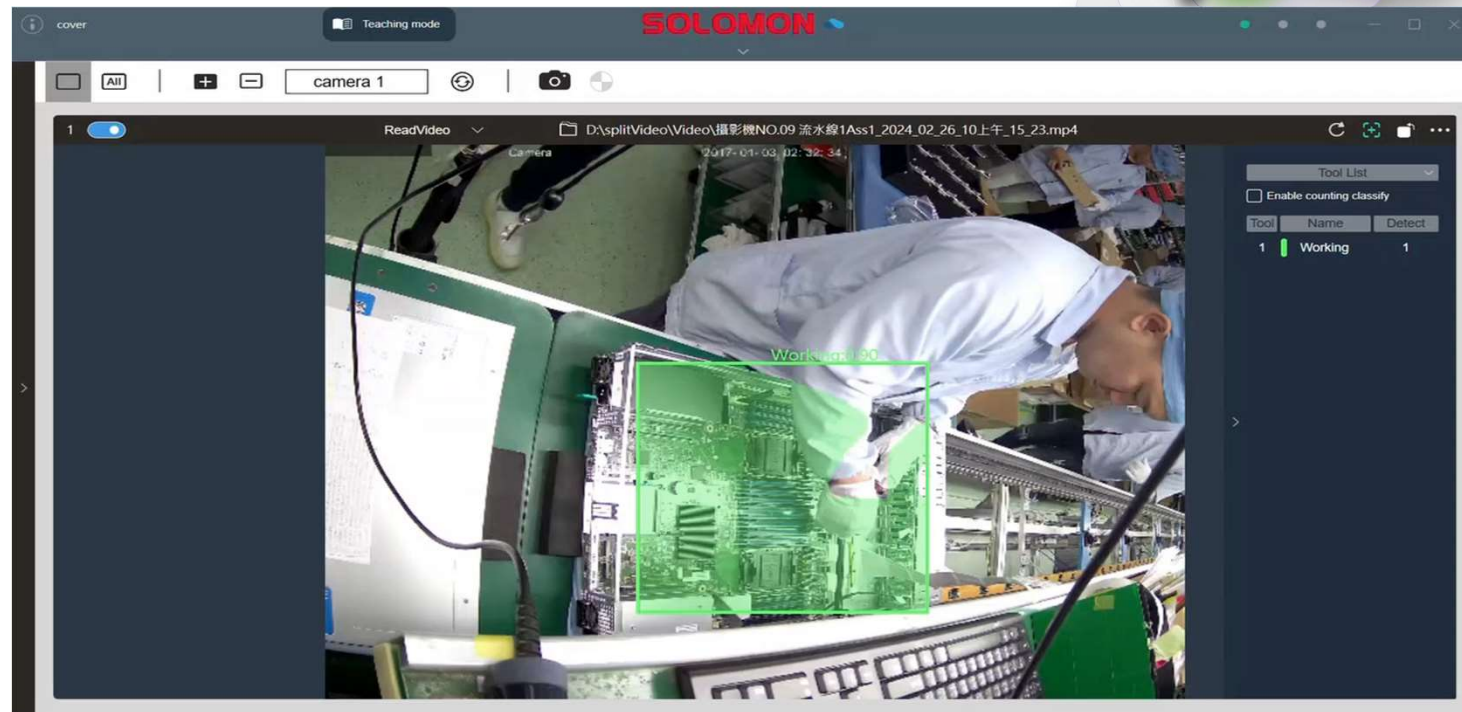
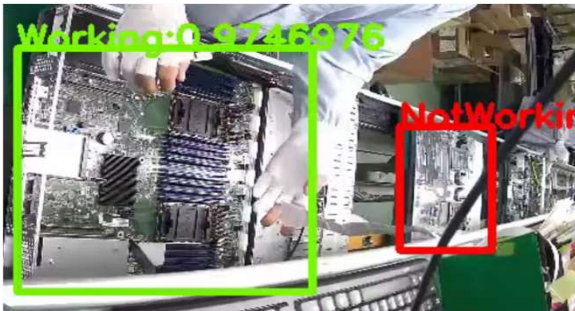
Defect Inspection for Optical Fiber Module



➡ We could do defect inspection on minor defect and welding point, through images captured with microscope

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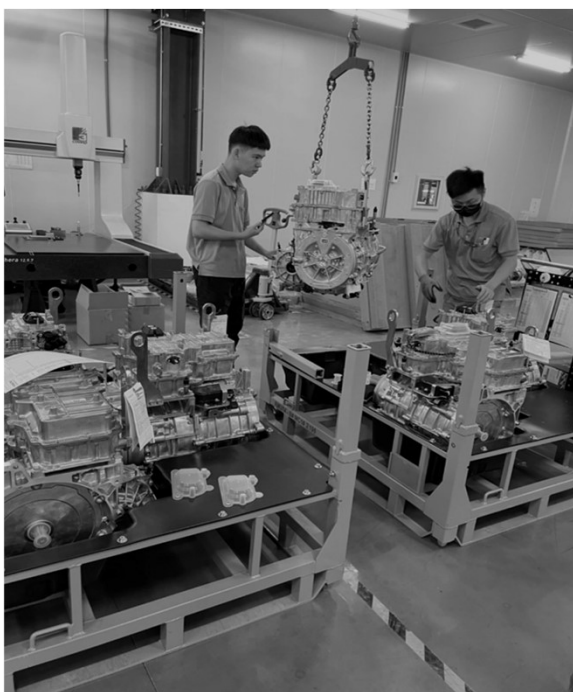
Assembly Station Time Study



→ SOLOMON AI monitors the work time at each assembly station to ensure work efficiency, track product status, and retain assembly records.

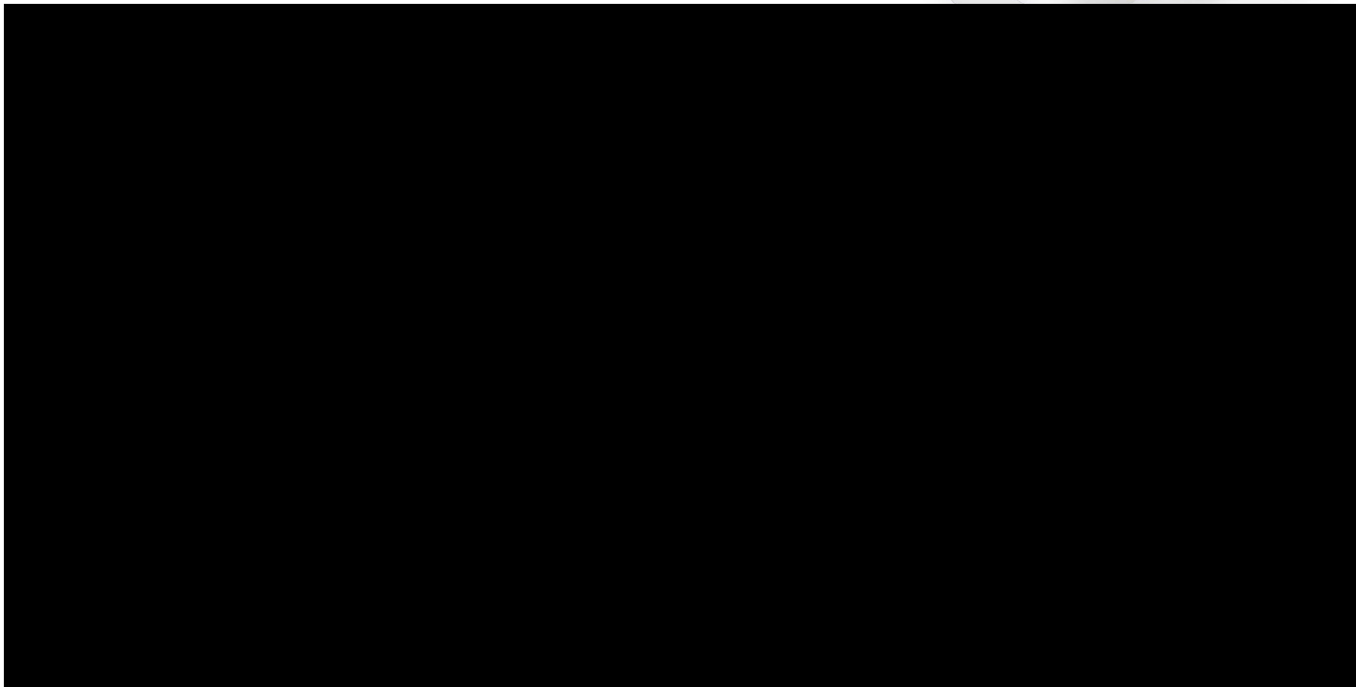
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Automotive Engine Inspection



Before
Around 8mins

32x Faster



After
Around 15secs

Conclusion

Breaking Limits: Powering the AI Era.



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Humanoid Robot With Intelligent Vision

